

# Gunter & Scott 1990 — Inventory of Definitions and Theorems

Source: **C. A. Gunter and D. S. Scott**, “**Semantic Domains**,” *Handbook of Theoretical Computer Science* Vol. B, North-Holland, 1990, pp. 633–674 ([../papers/Gunter Scott 1990.pdf](#)).

The work list for the Lean formalization: every definition and every one of the paper’s **30 numbered results** (Theorems / Lemmas / Proposition 1–30), in paper order, matched to its Lean equivalent.

## Progress (as of r0027, 2026-0806)

| #  | Quantity                              | Done                                                                        | Remaining                                  | Of        |
|----|---------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------|-----------|
| 1  | Definitions to define                 | <b>10</b>                                                                   | 3                                          | ≈13       |
| 2  | <b>Numbered</b> results complete      | <b>9</b> (Thm 1, Thm 3, Lem 4, Lem 5, Thm 6, Thm 7, Lem 8, Prop 15, Lem 19) | 20                                         | <b>29</b> |
| 2a | — of which <b>partially</b> proved    | Lem 10 (5 of 6 conjuncts — + unstated), Lem 17 (3 of 5 — ⊗ and + unstated)  | —                                          | —         |
| 3  | <b>Unnumbered prose claims</b> proved | <b>12</b>                                                                   | —                                          | —         |
| 4  | Mathlib foundations reused            | 12                                                                          | —                                          | 12        |
| 5  | Theorems in the development           | <b>199</b> live (+6 commented out as unused)                                | —                                          | —         |
| 6  | sorry in the development              | —                                                                           | <b>1</b> : thm18 in Skeleton/Section6.lean | —         |

Row 5 counts lines matching `^(@[...])?(theorem|lemma)` across the 27 modules.

**The sorry burn-down (from r0026).** The sorrys are deliberate scaffolding: fixed *statements* of outstanding results, confined to `ScottDomains/Skeleton/`, one file per agent worktree, so that three agents can prove them in parallel without any agent editing a declaration another depends on. Every other module remains sorry-free, and the count above is the burn-down metric — it goes  $10 \rightarrow 0$ . Round r0027, run as three agents in parallel, took it **10** → **1**.

| # | Open statement | Result                                                      | State after r0027                                                                                         |
|---|----------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1 | prop15         | Prop 15 — every bounded complete domain is bifinite         | <b>proved</b>                                                                                             |
| 2 | thm18          | Thm 18 — $D, D \rightarrow D$ domains $\implies D$ bifinite | <b>sorry</b> — the paper gives no proof, citing Smyth [Smy83a]; the obstacle is recorded in the docstring |
| 3 | lem19          | Lem 19 — the image of a closure is a domain                 | <b>proved</b> , via <code>IsClosure.rangeCompletePartialOrder</code>                                      |

| #    | Open statement                                       | Result                                                                                                                                                                                                                      | State after r0027                                                                                              |
|------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 4–7  | lem10_prod, lem10_smash,<br>lem10_lift, lem10_strict | Lem 10 —<br>bounded<br>completeness<br>closed under $\times, \otimes$ ,<br>( $\perp$ ), $\rightarrow \perp$ . The $\rightarrow$<br>conjunct is<br><b>already proved</b><br>(Thm 7’s<br>bounded-<br>complete half,<br>r0007) | <b>all four proved</b> ; Lem 10 is then <b>5 of 6</b> conjuncts<br>— $D + E$ is not stated                     |
| 8–10 | lem17_prod, lem17_lift, lem17_fun                    | Lem 17 —<br>bifiniteness<br>closed under $\times$ ,<br>( $\perp$ ), $\rightarrow$                                                                                                                                           | <b>all three proved</b> ; Lem 17 is then <b>3 of 5</b><br>conjuncts — $D \otimes E$ and $D + E$ are not stated |

**Why + is unstated in both.** ScottDomains/CoalescedSum.lean is 181 lines of ingredients — NonBotSum, sumBase, the same-side lemmas, sSup\_leftParts\_ne\_bot — but it defines **no sSup and no CompletePartialOrder instance**, and it is not imported by ScottDomains.lean. There is no cpo  $D + E$  to state a conjunct over. Finishing it is the prerequisite for lem10\_sum and lem17\_sum, and its sSup must branch on landing in NonBotSum, not on directedness — the defect fixed in ScottHom and then in Smash would otherwise recur a third time. lem17\_smash is unstated for no such reason: it was simply left out of the r0026 skeleton.

**The smashSup defect (r0027).** lem10\_smash was not merely open: as smashSup stood, it was **false**, and the kernel confirmed a refutation. smashSup branched its dite on the base being nonempty *and directed*, so a merely **bounded** non-directed base fell to the adjoined  $\perp$ , which is not even an upper bound. Witness:  $D = \text{Prop} \times \text{Prop}$ ,  $E = \text{Prop}$ ,  $s = \{\uparrow((\text{True}, \text{False}), \text{True}), \uparrow((\text{False}, \text{True}), \text{True})\}$ , bounded by  $\uparrow((\text{True}, \text{True}), \text{True})$ . This is the same defect ScottHom.lean records having already hit for the function space. The repair branches on the coordinatewise supremum landing in NonBotPair — the condition under which it is an element of  $D \otimes E$  at all, rather than a merely sufficient condition for it. smashSup\_of\_directed and smashSup\_of\_empty kept their statements and were reproved, which is the agreement claim stated in Lean rather than in prose, and smashCpo needed no change.

Kernel check on the r0027 merges (#print axioms): all ten proved statements plus smashCpo, smashSup\_of\_directed and smashSup\_of\_empty depend only on propext, Classical.choice, Quot.sound — lem10\_prod and lem19 do not even need Classical.choice. None depends on sorryAx. thm18 does, as its sorry requires.

Row 2 counts only the paper’s 30 **numbered** results (Theorems / Lemmas / Proposition 1–30). Row 3 counts the claims the paper makes **in prose** rather than as numbered results; these are paper content too, and all twelve are formally verified:

| # | Paper claim                                                                        | Where       | Lean                                                               |
|---|------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------|
| 1 | “the compact elements [of $P \rightarrow N$ ] are just the finite subsets of $N$ ” | p. 9        | isCompactElement_iff_finite                                        |
| 2 | “ $P \rightarrow N$ ... is a domain”                                               | p. 9        | instance :<br>Domain (Set X)                                       |
| 3 | “ $D \rightarrow E$ is a ... cpo”                                                  | Thm 7 proof | instance :<br>CompletePartialOrder<br>(ScottHom $\alpha$ $\beta$ ) |
| 4 | “... bounded complete ... whenever $E$ is”                                         | Thm 7 proof | instance :<br>BoundedComplete<br>(ScottHom $\alpha$ $\beta$ )      |
| 5 | “step(s) ... is continuous”                                                        | Thm 7 proof | scottContinuous_stepFun                                            |
| 6 | “... and compact in the ordering on $D \rightarrow E$ ”                            | Thm 7 proof | isCompactElement_step                                              |
| 7 | “they form a basis for $D \rightarrow E$ ”                                         | Thm 7 proof | instance :<br>IsAlgebraic<br>(ScottHom $\alpha$ $\beta$ )          |

| #  | Paper claim                                                              | Where                 | Lean                                     |
|----|--------------------------------------------------------------------------|-----------------------|------------------------------------------|
| 8  | every compact function is a <i>finite</i> join of step functions         | Thm 7 proof, implicit | <code>exists_finite_isLUB_of_isCo</code> |
| 9  | “an embedding is an injection”                                           | §3.1                  | <code>IsEmbeddingProjectionPair.i</code> |
| 10 | “a projection is a surjection”                                           | §3.1                  | <code>IsEmbeddingProjectionPair.s</code> |
| 11 | “it is easy to check that $p_N \dots$ is a finitary projection”          | §3.1                  | <code>isFinitaryProjection_normal</code> |
| 12 | “the set of strict continuous functions $D \rightarrow E$ is also a cpo” | §2.1                  | <code>ScottDomains.strictHomCpo</code>   |

Six of those eleven are the body of **Theorem 7**, which is now **complete** — all four conjuncts of its conclusion (cpo, bounded complete, algebraic, countably based) are formally verified, as `ScottHom.isBoundedCompleteDomain_scottHom`.

It is proved under **weaker hypotheses than the paper states**: bounded completeness of  $D$  is never used.  $D$  need only be a domain. The function space is a cpo for any preordered  $D$ , algebraic when  $D$  and  $E$  are, bounded complete because  $E$  is, and countably based because  $D$  and  $E$  are.

The remaining theorems are supporting API: the  $\ll$  calculus, the `compactsBelow` machinery, the pointwise order and suprema on  $D \rightarrow E$ , and the step-function adjunction. The paper assumes or elides all of it.

**Reference audit (r0020).** Of the theorems in the development, 16 are never cited elsewhere. Nine of those are *terminal by design* — they are the paper’s own claims, so nothing should cite them (`injective_embedding`, `surjective_projection`, the `isNormalIn_sUnion*` family and `mono_right` for Lemma 4, `singleton_bot_isNormalIn` for  $\{\perp\} \in P(C)$ , `isLeast_kleeneFix_le`, `eq_kleeneOperator_op`). The other six were speculative API written for callers that never appeared; they are **commented out in place**, each with a note on why it exists and what is instructive about it, and the build is unchanged — which also confirms that the three `@[simp]` ones among them were never firing implicitly.

The development is **27 modules, 4440 lines, 1 sorry, 0 other warnings**. Counts of definitions, results and theorems are in the Progress table above — they are not repeated here, so that this section cannot drift out of step with it. What each round delivered:

| # | Round       | Module                                             | Contents                                                                                                                                                                                                                                               |
|---|-------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | r0003       | <code>ScottDomains/WayBelow.lean</code>            | <code>way-below</code> $\ll$ at $[Preorder\ \alpha]$ , 7 theorems; $x \ll x \leftrightarrow IsCompactElement\ x$ holds by <code>Iff.rfl</code>                                                                                                         |
| 2 | r0004       | <code>ScottDomains/Domain.lean</code>              | <code>IsAlgebraic</code> , <code>Domain</code> (with the paper’s countable-basis condition), <code>BoundedComplete</code> , 8 theorems, <code>Domain Prop</code>                                                                                       |
| 3 | r0005       | <code>ScottDomains/Powerset.lean</code>            | the paper’s $P\ \mathbb{N}$ (p. 9): compacts of <code>Set X</code> are exactly the finite subsets, hence <code>Domain (Set ℕ)</code> — the nondegenerate witness                                                                                       |
| 4 | r0006–r0007 | <code>ScottDomains/ScottHom.lean</code>            | the continuous function space $D \rightarrow E$ : <code>ScottHom</code> , the pointwise order, <code>CompletePartialOrder</code> , and <code>BoundedComplete</code> when $E$ is — <b>Theorem 7’s first sentence in full</b>                            |
| 5 | r0008       | <code>ScottDomains/StepFunction.lean</code>        | the single step function <code>step k e</code> : continuity (from $k$ compact), the adjunction <code>step k e ≤ f ↔ e ≤ f k</code> , and compactness in $D \rightarrow E$ (from $e$ compact)                                                           |
| 6 | r0009       | <code>ScottDomains/FunctionSpaceDomain.lean</code> | <code>IsAlgebraic</code> — the paper’s “they form a basis for $D \rightarrow E$ ”; the two halves of <code>IsAlgebraic</code> use disjoint hypotheses (directedness needs only $E$ bounded complete; the <code>lub</code> needs only $D, E$ algebraic) |

| #  | Round       | Module                                     | Contents                                                                                                                                                                                                 |
|----|-------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7  | r0010       | ScottDomains/CompactFunctionSpace.lean     | <b>lean</b> compact function is a <b>finite</b> join of step functions — the finiteness half of the basis claim                                                                                          |
| 8  | r0011       | ScottDomains/FunctionSpaceCompactable.lean | <b>Table 1</b> <b>lean</b> uncountable, hence Domain (ScottHom $\alpha$ $\beta$ ) — <b>Theorem 7 complete</b>                                                                                            |
| 9  | r0012       | ScottDomains/NormalSubposet.lean           | <b>lean</b> normal-subposet relation $\triangleleft$ and <b>Lemma 4</b> , all four parts                                                                                                                 |
| 10 | r0012–r0013 | ScottDomains/Projection.lean               | embedding–projection pairs, projections, the paper’s “an embedding is an injection, a projection is a surjection”; <b>im(p) is a cpo</b> and <b>finitary projections</b>                                 |
| 11 | r0014       | ScottDomains/FinitaryProjection.lean       | <b>Lemma 5</b> — the compacts of $\text{im}(p)$ are $\text{im}(p) \cap K(D)$ (needs only that $p$ is a projection), and $\text{im}(p) \cap K(D) \triangleleft K(D)$                                      |
| 12 | r0015       | ScottDomains/NormalProjection.lean         | <b>p_N</b> $= \bigsqcup \{y \in N \mid y \sqsubseteq x\}$ : continuous, a projection, and $\text{im}(p_N) \cap K(D) = N$ — half of <b>Theorem 6</b> ’s correspondence, plus order preservation both ways |
| 13 | r0016       | ScottDomains/Theorem6.lean                 | <b>Theorem 6</b> — $p_N$ is finitary, $p_{\{\text{im}(p) \cap K(D)\}} = p$ , and the correspondence assembled                                                                                            |
| 14 | r0017       | ScottDomains/FixedPoint.lean               | <b>Theorem 1</b> — $\bigsqcup_n f^n(\perp)$ is the least fixed point of a continuous $f$ on a cpo. Not Mathlib reuse; see the §2 table                                                                   |
| 15 | r0018       | ScottDomains/UniformFixedPoint.lean        | <b>Theorem 3</b> — $\text{fix}$ is the unique uniform fixed-point operator; $\downarrow a$ as a cpo                                                                                                      |
| 16 | r0019       | ScottDomains/Product.lean                  | $D \times E$ as a cpo (the one construction needing no case split) and <b>Lemma 8 parts 1–3</b>                                                                                                          |
| 17 | r0021       | ScottDomains/Currying.lean                 | <b>Lemma 8.4</b> — currying, $D \rightarrow (E \rightarrow F) \cong (D \times E) \rightarrow F$ , and with it <b>Lemma 8 complete</b>                                                                    |
| 18 | r0022       | ScottDomains/EffectivePresentation.lean    | <b>§4.2’s effective presentation</b> — the enumeration of the basis with its two decidability conditions                                                                                                 |
| 19 | r0023       | ScottDomains/Lift.lean                     | the <b>lift</b> $D_1$ as a cpo, on Mathlib’s WithBot                                                                                                                                                     |
| 20 | r0024       | ScottDomains/StrictHom.lean                | the <b>strict function space</b> $D \rightarrow_1 E$ as a cpo — needs no case split, since both branches of ScottHom’s $s\text{Sup}$ are strict                                                          |
| 21 | r0025       | ScottDomains/Smash.lean                    | §4.3’s <b>smash product</b> $D \otimes E$ — the non-bottom pairs with a new bottom adjoined, as a cpo                                                                                                    |
| 22 | r0025       | ScottDomains/Bifinite.lean                 | §6.1’s <b>Plotkin order</b> and <b>bifinite</b> domain                                                                                                                                                   |

**§2 and §3 are now complete** — the only §3 omission is the paper’s *computable function*, which needs an r.e.-predicate notion Mathlib does not supply.

#### Dependency structure of the remaining definitions

| # | Definition                  | Prerequisites    | Status        |
|---|-----------------------------|------------------|---------------|
| 1 | smash product $D \otimes E$ | Domain, OrderBot | ✓ done, r0025 |

| # | Definition               | Prerequisites                                               | Status                                                                  |
|---|--------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------|
| 2 | bifinite / Plotkin order | IsNormalIn ( $\leq$ , r0012)                                | ✓ <b>done, r0025</b>                                                    |
| 3 | sum $D + E$              | Domain                                                      | independent, ready                                                      |
| 4 | $D_\infty$               | embedding–projection pairs (r0012)                          | independent, but large (inverse limits)                                 |
| 5 | the three powerdomains   | the <b>ideal completion</b> (Theorem 11) — <i>not</i> built | <b>blocked</b> ; the three are independent of each other once it exists |

Rows 1 and 2 were written as two modules and checked in a single build, which is the parallelism the structure actually admits — the bottleneck is the build-and-fix loop, not the writing.

Next: the sum  $D + E$ , then Theorem 11 to unblock the powerdomains, then Lemma 9 (the strict analogues of Lemma 8) and Lemma 10 (closure of bounded completeness).

## Work counts

- **Reuse from Mathlib — no work (11):** poset, directed, cpo,  $\perp$ , monotone, continuous, fixed-point operator, Schröder–Bernstein (2), algebraic lattice, product,  $\lambda$ -notation. (Theorem 1 was listed here in an earlier draft and has been removed: Mathlib’s `OrderHom.lfp` is Knaster–Tarski over a complete lattice, not Kleene’s  $\bigsqcup_n f^n(\perp)$  over a cpo. It is proved in `ScottDomains/FixedPoint.lean`, so **29** numbered results needed proof, not 28.)
- **Generalize / adapt (4):** compact element (`IsCompactElement` — its *definition* is already stated at `[PartialOrder  $\alpha$ ]` and so applies to a dcpo verbatim; what is `CompleteLattice`-only is every lemma about it, so the dcpo API must be re-proved), sum, lift (`WithBot/Sum`), ideal completion (`Order.Ideal`).
- **Definitions to define — new ( $\approx 13$ ; 4 done in r0003–r0004, 9 remaining):** way-below  $\ll$  (**done** — `ScottDomains/WayBelow.lean`), algebraic cpo, **domain**, bounded-complete (**all three done** — `ScottDomains/Domain.lean`), embedding–projection pair, (finitary) projection, normal subposet, effective presentation, smash product, the three powerdomains (Hoare / Smyth / Plotkin), bifinite / Plotkin order, and  $D_\infty$ .
- **Theorems to prove (28):** 28 of the paper’s 30 numbered results — Theorems 3, 6, 7, 11, 12, 14, 16, 18, 21, 22, 25, 26, 27, 29; Lemmas 4, 5, 8–10, 13, 17, 19, 20, 23, 24, 28, 30; Proposition 15. Only Theorems 1 & 2 come free from Mathlib.

**Bottom line:**  $\approx 13$  definitions to define + 28 results to prove, on top of 12 reused Mathlib foundations. After r0004: **9 definitions + 28 results** remain.

**Lean column legend:** ✓ reuse Mathlib (name given) · ~ partial (Mathlib has a related or lattice-only version — generalize) · ✕ define / prove (absent from Mathlib v4.32.2, confirmed by grep).

Statements are paraphrased/de-garbled from the PDF (1990 Type-3 fonts render → as !, drop fi ligatures, mangle math), so read them as a guide to *what* each result is. Notation:  $\sqsubseteq$  order,  $\bigsqcup$  directed sup,  $\perp$  bottom,  $\ll$  way-below,  $K(D)$  compacts,  $Fp(D)$  finitary projections.

## §2 Recursive definitions of functions

| §   | Kind | Ref | Concept / statement                                                                 | In Lean / Mathlib?                                                                    |
|-----|------|-----|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 2.1 | Def  | —   | <b>poset</b> (partially ordered set)                                                | ✓ <code>PartialOrder</code>                                                           |
| 2.1 | Def  | —   | <b>directed</b> subset $M$ : every finite $u \subseteq M$ has an upper bound in $M$ | ✓ <code>DirectedOn / IsDirected</code>                                                |
| 2.1 | Def  | —   | <b>cpo</b> : poset in which every directed $M$ has a lub $\bigsqcup M$              | ✓ <code>CompletePartialOrder</code> (chains: <code>OmegaCompletePartialOrder</code> ) |
| 2.1 | Def  | —   | <b>bottom</b> $\perp$ (least element)                                               | ✓ <code>OrderBot</code>                                                               |
| 2.1 | Def  | —   | <b>monotone</b> function                                                            | ✓ <code>Monotone / OrderHom</code>                                                    |
| 2.1 | Def  | —   | <b>continuous</b> : monotone, $f(\bigsqcup M) = \bigsqcup f(M)$ for directed $M$    | ✓ <code>ScottContinuous</code>                                                        |

| §   | Kind | Ref | Concept / statement                                                                                                | In Lean / Mathlib?                                                                                                                                                                      |
|-----|------|-----|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.1 | Thm  | 1   | <b>Fixed-Point Theorem:</b> $f : D \rightarrow D$ continuous $\implies$ least fixed point $\bigsqcup_n f^n(\perp)$ | ✓ <b>proved</b> (r0017) —<br>ScottDomains.theorem1. <b>Not</b> Mathlib reuse: OrderHom.lfp is Knaster–Tarski over a <i>complete lattice</i> with only monotonicity, a different theorem |
| 2.2 | Thm  | 2   | <b>Schröder–Bernstein</b> for sets                                                                                 | ✓ Function.schroeder_bernstein (SetTheory/Cardinal/SchroederBernstein.lean — the name in an earlier draft of this row, Function.Embedding.schroederBernstein, does not exist            |
| 2.3 | Def  | —   | <b>fixed-point operator</b> (uniform)                                                                              | ✓ OrderHom.lfp / LawfulFix                                                                                                                                                              |
| 2.3 | Thm  | 3   | The standard operator is the <b>unique uniform</b> fixed-point operator                                            | ✓ <b>proved</b> (r0018) —<br>ScottDomains.theorem3; a fixed point operator is formalized as a family over every CompletePartialOrder in a universe                                      |

### §3 Effectively presented domains

| §   | Kind | Ref | Concept / statement                                                                                                                                                                                                                  | In Lean / Mathlib?                                                                                                                                                   |
|-----|------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.1 | Def  | —   | <b>compact element</b> $x: x \sqsubseteq \bigsqcup M(\text{dir.}) \implies x \sqsubseteq y$ some $y \in M; K(D)$                                                                                                                     | ~ IsCompactElement — def is [PartialOrder], so reusable on a dcpo; its lemmas are all CompleteLattice-only                                                           |
| 3.1 | Def  | —   | <b>way-below</b> $\ll$ (approximation relation behind compactness)                                                                                                                                                                   | ✓ ScottDomains.WayBelow (r0003) —<br>$x \ll x \leftrightarrow \text{IsCompactElement } x$ by Iff.rfl                                                                 |
| 3.1 | Def  | —   | <b>algebraic</b> cpo: $x = \bigsqcup \{x' \in K(D) : x' \sqsubseteq x\}$ (directed)                                                                                                                                                  | ✓ ScottDomains.IsAlgebraic (r0004)                                                                                                                                   |
| 3.1 | Def  | —   | <b>domain</b> = algebraic cpo <b>whose basis <math>K(D)</math> is countable</b> (the paper’s definition, p. 9 — the countability condition was missing from an earlier draft of this row)                                            | ✓ ScottDomains.Domain (r0004)                                                                                                                                        |
| 3.1 | Def  | —   | <b>bounded complete:</b> $\perp +$ every bounded subset has a sup — a <i>separate</i> predicate; the paper composes them as “bounded complete domain” (Thm 7, Lem 10, Lem 13, Thm 14), which is the literature’s <i>Scott domain</i> | ✓ ScottDomains.BoundedComplete (r0004); the compound is [Domain $\alpha$ ] [BoundedComplete $\alpha$ ]                                                               |
| 3.1 | Def  | —   | <b>(countably based) algebraic lattice</b>                                                                                                                                                                                           | ✓ IsCompactlyGenerated (+CompleteLattice)                                                                                                                            |
| 3.1 | Def  | —   | <b>embedding–projection pair</b> $(g, f)$                                                                                                                                                                                            | ✓<br>ScottHom.IsEmbeddingProjectionPair (r0012)                                                                                                                      |
| 3.1 | Def  | —   | <b>projection; finitary projection</b> $p$ : $p \circ p = p \sqsubseteq \text{id}$ , $\text{im}(p)$ a domain                                                                                                                         | ✓ ScottHom.IsProjection (r0012),<br>ScottHom.IsFinitaryProjection (r0013) — $\text{im}(p)$ carries a CompletePartialOrder via IsProjection.rangeCompletePartialOrder |
| 3.1 | Def  | —   | <b>normal subposet</b> / substructure                                                                                                                                                                                                | ✓ ScottDomains.IsNormalIn, notation $\triangleleft$ (r0012)                                                                                                          |

| §   | Kind | Ref | Concept / statement                                                                                                                          | In Lean / Mathlib?                                                                                                                                                                                                                                                                                 |
|-----|------|-----|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.1 | Lem  | 4   | $\langle P(C), \triangleleft \rangle$ of substructures is a cpo with $\{\perp\}$ least                                                       | ✓ <b>proved</b> (r0012), all four parts                                                                                                                                                                                                                                                            |
| 3.1 | Lem  | 5   | $p$ finitary projection $\implies$ compacts of $\text{im}(p)$ are $\text{im}(p) \cap K(D)$ , and $\text{im}(p) \cap K(D) \triangleleft K(D)$ | ✓ <b>proved</b> (r0014) — <code>IsProjection.isCompactElement_iff</code> (needs only <i>projection</i> ) and <code>IsFinitaryProjection.isNormalIn_compacts</code>                                                                                                                                 |
| 3.1 | Thm  | 6   | Isomorphism: normal substructures $\cong \text{Fp}(D)$ (finitary projections)                                                                | ✓ <b>proved</b> (r0015–r0016) — <code>ScottDomains.theorem6</code>                                                                                                                                                                                                                                 |
| 3.1 | Thm  | 7   | $D, E$ bounded-complete domains $\implies D \rightarrow E$ bounded-complete domain                                                           | ✓ <b>proved</b> (r0006–r0011) — <code>ScottHom.isBoundedCompleteDomain_scottHom</code> ; $D$ bounded complete is not needed                                                                                                                                                                        |
| 3.2 | Def  | —   | <b>effective presentation</b> $d : \mathbb{N} \rightarrow K(D)$ ; <b>effectively presented domain</b>                                        | ✓ <code>ScottDomains.EffectivePresentation</code> (r0022). The paper’s <b>computable function</b> is <i>not</i> formalized: it needs “recursively enumerable”, and Mathlib v4.32.2 has no <code>RePred</code> or equivalent (grep finds none) — building it is recursion theory, not domain theory |

## §4 Operators and functions

| §   | Kind | Ref | Concept / statement                                                                                                                                                                                                                      | In Lean / Mathlib?                                                                                                                                                                                          |
|-----|------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.1 | Def  | —   | <b>product</b> $D \times E$                                                                                                                                                                                                              | ✓ Prod order from Mathlib; the <b>cpo instance</b> is <code>ScottDomains.instCompletePartialOrderProd</code> (r0019) — Mathlib has <code>Prod.supSet</code> and <code>isLUB_prod</code> but no cpo instance |
| 4.2 | Def  | —   | <b>Church’s <math>\lambda</math>-notation</b> (continuous abstraction)                                                                                                                                                                   | ✓ <code>OrderHom / <math>\omega</math>CPO ContinuousHom</code>                                                                                                                                              |
| 4.3 | Def  | —   | <b>smash product</b> $D \otimes E$                                                                                                                                                                                                       | ✓ <code>ScottDomains.Smash + smashCpo</code> (r0025)                                                                                                                                                        |
| 4.4 | Def  | —   | <b>sum</b> $D + E$ ; <b>lift</b> $D \perp$                                                                                                                                                                                               | lift ✓ <code>ScottDomains.liftCpo</code> on <code>WithBot</code> (r0023) — Mathlib gives the order, not the cpo; sum still ~                                                                                |
| 4.x | Lem  | 8   | $D \times E \cong E \times D$ ; $(D \times E) \times F \cong D \times (E \times F)$ ; $D \rightarrow (E \times F) \cong (D \rightarrow E) \times (D \rightarrow F)$ ; $D \rightarrow (E \rightarrow F) \cong (D \times E) \rightarrow F$ | ✓ <b>proved</b> — <code>prodComm</code> , <code>prodAssoc</code> , <code>scottHomProd</code> (r0019); <code>scottHomCurry</code> (r0021)                                                                    |
| 4.x | Lem  | 9   | Product/function-space iso laws over $D, E, F$                                                                                                                                                                                           | ✗ prove                                                                                                                                                                                                     |
| 4.5 | Lem  | 10  | $D, E$ bounded complete $\implies \rightarrow, \times, \otimes, +, () \perp$ bounded complete                                                                                                                                            | ✗ prove                                                                                                                                                                                                     |
| 4.5 | Thm  | 11  | <b>Ideal completion</b> of a countable pre-order is a domain (all domains so arise)                                                                                                                                                      | ~ <code>Order.Ideal</code> exists $\rightarrow$ prove                                                                                                                                                       |
| 4.5 | Thm  | 12  | Initiality of a continuous algebra satisfying axioms T                                                                                                                                                                                   | ✗ prove                                                                                                                                                                                                     |
| 4.5 | Lem  | 13  | $D$ bounded complete $\implies$ powerdomains $D], D[$ bounded complete                                                                                                                                                                   | ✗ prove                                                                                                                                                                                                     |
| 4.5 | Thm  | 14  | Equivalent characterizations of an (algebraic/BC) domain                                                                                                                                                                                 | ✗ prove                                                                                                                                                                                                     |

## §5 Powerdomains

| §   | Kind | Ref | Concept / statement                                                | In Lean / Mathlib? |
|-----|------|-----|--------------------------------------------------------------------|--------------------|
| 5.1 | Def  | —   | <b>powerdomain</b> (non-deterministic outcomes)                    | ✗ define           |
| 5.2 | Def  | —   | <b>Hoare (lower), Smyth (upper), Plotkin (convex)</b> powerdomains | ✗ define           |
| 5.3 | —    | —   | Universal & closure properties (see Lem 13, 28, 30)                | ✗ prove            |

## §6 Bifinite (SFP) domains

| §   | Kind | Ref | Concept / statement                                                                                        | In Lean / Mathlib?                                             |
|-----|------|-----|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 6.1 | Def  | —   | <b>Plotkin order / bifinite (SFP) domain</b>                                                               | ✓ <code>ScottDomains.IsPlotkinOrder, IsBifinite (r0025)</code> |
| 6.1 | Prop | 15  | Every bounded-complete domain is bifinite                                                                  | ✗ prove                                                        |
| 6.1 | Thm  | 16  | $D \text{ bifinite} \implies \text{Fp}(D)$ is an algebraic lattice                                         | ✗ prove                                                        |
| 6.2 | Lem  | 17  | $D, E \text{ bifinite} \implies \rightarrow, \times, \otimes, +, ()^\perp$ bifinite (incl. function space) | ✗ prove                                                        |
| 6.2 | Thm  | 18  | If $D$ and $D \rightarrow D$ are domains, then $D$ is bifinite                                             | ✗ prove                                                        |
| 6.2 | Lem  | 19  | closure $r : D \rightarrow D$ ( $r \circ r = r \sqcup \text{id}$ ) $\implies \text{im}(r)$ is a domain     | ✗ prove                                                        |
| 6.2 | Lem  | 20  | $D \text{ domain} \implies \text{Fc}(D)$ (finitary closures) is a cpo                                      | ✗ prove                                                        |

## §7 Recursive definitions of domains (universal domain, $D_\infty$ )

| § | Kind | Ref | Concept / statement                                                                                                   | In Lean / Mathlib?                                   |
|---|------|-----|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 7 | Def  | —   | <b>recursive domain equation; universal domain</b> / $D_\infty$ (inverse limit)                                       | ✗ define                                             |
| 7 | Thm  | 21  | $F$ representable over cpo $U \implies$ a domain $D$ with $D \cong F(D)$                                              | ✗ prove                                              |
| 7 | Thm  | 22  | Any countably-based algebraic lattice $L$ : a closure $r : P(\mathbb{N}) \rightarrow L$                               | $\sim \text{IsCompactlyGenerated} \rightarrow$ prove |
| 7 | Lem  | 23  | The function-space operator is representable over $P(\mathbb{N})$                                                     | ✗ prove                                              |
| 7 | Lem  | 24  | $U$ cpo; $\times$ and $\rightarrow$ representable over $U \implies$ (setup for universality)                          | ✗ prove                                              |
| 7 | Thm  | 25  | $U$ non-trivial domain representing $\times, \rightarrow \implies U$ <b>universal</b>                                 | ✗ prove                                              |
| 7 | Thm  | 26  | Any signature $(s_1, \dots, s_n)$ : combinators $F_1, \dots, F_n$ solving the equations                               | ✗ prove                                              |
| 7 | Thm  | 27  | Any bounded-complete $D$ : a projection of the universal domain onto $D$                                              | ✗ prove                                              |
| 7 | Lem  | 28  | Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()^\perp, ()^\perp$ representable over $U$                      | ✗ prove                                              |
| 7 | Thm  | 29  | $D \text{ bifinite} \implies D^+ \text{ bifinite}$ ; solving $D \cong D^+$                                            | ✗ prove                                              |
| 7 | Lem  | 30  | Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()^\perp, ()^\perp$ p-representable over universal bifinite $V$ | ✗ prove                                              |



**Tally (matched):** ✓ reuse Mathlib  $\approx$  **12** (poset, directed, cpo,  $\perp$ , monotone, continuous, fixed-point Thm 1 & operator, Schröder–Bernstein, algebraic lattice, product,  $\lambda$ -notation) ·  $\sim$  partial  $\approx$  **4** (compact element, sum/lift, ideal completion Thm 11, Thm 22) ·  $\times$  define / prove  $\approx$  **44**, of which 4 are done ( $\ll$  r0003; algebraic, domain, bounded-complete r0004)  $\rightarrow$  **40 remaining** — the bifinite / powerdomain /  $D^\infty$  development and all 28 numbered results.

What is done, and what is next, is listed under **Progress** at the top of this file.